



Longitudinal relations among expectancy, mastery-approach, effort, and achievement

Jianzhong Xu¹

Received: 11 November 2024 / Accepted: 5 May 2025 / Published online: 8 June 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

This study aims to explore longitudinal associations among homework expectancy, mastery-approach, homework effort, and mathematics achievement involving middle school students over one school year. Results indicated that there was support for a positive reciprocal influence of mastery-approach and effort; greater initial mastery-approach resulted in greater later effort, and higher initial effort resulted in higher later mastery-approach. Results further indicated that higher initial mastery-approach led to higher later expectancy and achievement. Furthermore, the latent expectancy $\times$ mastery-approach interaction showed a significant effect on subsequent achievement; mastery-approach exerted a more positive effect on subsequent achievement when aligned with higher expectancy.

Keywords Achievement · Effort · Expectancy · Homework · Mastery-approach · Middle school

1 Introduction

According to expectancy-value theory (Eccles & Wigfield, 2020, 2023; Wigfield et al., 2015) and achievement goal theory (Cellar et al., 2011; Guo et al., 2023; Urdan & Kaplan, 2020), expectancy, mastery-approach, effort, and achievement are inextricably interconnected, influencing each other over time (Abaszadeh et al., 2024; Diseth, 2011; Howard et al., 2021; Kaplan & Flum, 2010; King et al., 2024; Pekrun et al., 2006; Tamura et al., 2022). However, as will be discussed later, much of the existing research on these associations relies on cross-sectional studies or studies that fail to control for prior levels of these constructs (Howard et al., 2021; Pekrun et al., 2006; Tamura et al., 2022). Considering the inherent interplay

✉ Jianzhong Xu
jx18@colled.msstate.edu

¹ Department of Counseling, Higher Education Leadership, Educational Psychology, and Foundations, Mississippi State University, P.O. Box 9727, Mississippi State, MS 39762, USA

between these constructs, it is crucial to utilize cross-lagged models to explore their longitudinal associations. Whereas such models do not determine causation, they allow for a more rigorous analysis of temporal dynamics, enhancing empirical foundation underpinning causal inference (Marsh et al., 2016; Xu, 2025a).

While cross-lagged panel models can be used to examine the longitudinal relationships among these constructs across different domains (e.g., foreign language) and development stages, middle school mathematics homework was the focus of our study for the following considerations. First, as a ubiquitous educational practice across nations, homework is a daily routine that significantly influence students, families, and teachers (Corno & Xu, 2004; Fernández-Alonso et al., 2015; Xu et al., 2024). Second, research on homework underscores the necessity of adopting a domain-specific perspective (Trautwein et al., 2006; Xu, 2024b). Third, mathematics holds global significance owing to its crucial role in STEM areas (Kennedy & Odell, 2023; Wang, 2013) and often demands greater effort and more homework from students compared to other subjects (Marsh et al., 2016; Silinskas & Kikas, 2019; Xu, 2022, 2024a). Yet, early adolescence is marked by a notable decline in expectancy and mastery-approach (Eccles & Wigfield, 2023; Tuominen et al., 2020; Wigfield et al., 2015; Xiang et al., 2022), making middle school mathematics homework a particularly relevant focus for investigation.

This line of investigation would be especially intriguing in non-Western countries like China. As in many East Asian countries shaped by Confucian traditions, Chinese culture underscores the importance of mastery-approach (e.g., growth and self-improvement), sustained effort (e.g., diligence and resilience), and academic success (e.g., long-term perspective on academic goals and pursuits; Chen & Wong, 2015; Cui et al., 2023; Eccles & Wigfield, 2020; Wang, 2018; Wigfield et al., 2015; Xu, 2023b). These cultural norms and expectations likely influence the longitudinal relationships among expectancy, mastery-approach, effort, and achievement.

1.1 Relationships among expectancy, mastery-approach, effort, and achievement

This study is grounded in two major theories: expectancy-value and achievement goal theories. Expectancy-value theory (Eccles & Wigfield, 2020, 2023; Eccles et al., 1983) is a widely recognized framework for understanding achievement-related behavior. It postulates that individuals' motivation to engage in an upcoming task is shaped by two major elements – expectancy of success and subjective task value. Expectancy refers to individuals' subjective assessment of their capacity to succeed in a task, addressing the question, “Can I do this task?” (Wigfield et al., 2015, p. 659). Value pertains to the importance individuals place on a task (intrinsic value, attainment value, utility value, and cost), addressing the question, “Do I want to do this task?” (Wigfield et al., 2015, p. 659).

While both expectancy and value positively correlate with a broad spectrum of adaptive achievement outcomes – such as choice, effort, persistence, and academic performance – a distinct pattern arises when they are examined together. Expectancy tends to be a stronger predictor of academic achievement, whereas

value is more closely linked to sustained interest and upcoming course enrollment (Barron & Hulleman, 2015; Urhahne & Wijnia, 2023; Eccles & Wigfield, 2020). Additionally, expectancy-value theory (Eccles & Wigfield, 2020; Eccles et al., 1983) postulates that prior achievement influence subsequence expectancy (e.g., through causal attribution); individuals with successful experiences in a particular field are more inclined to expect future success in that field.

Achievement goal theory is an overarching framework to comprehend the goals individuals strive in academic settings, encompassing mastery orientation and performance orientation (Cellar et al., 2011; Urdan & Kaplan, 2020; Wigfield & Cambria, 2010). Mastery orientation, driven by a desire for personal growth and self-improvement, emphasizes learning and understanding, posing questions such as “How can I do this task?” and “What will I learn?” (Wigfield & Cambria, 2010, p. 7). In contrast, performance orientation is motivated by the pursuit of external rewards and the desire of favorable judgements from others. It focuses on outperforming peers and demonstrating competence to external evaluators, addressing questions, “Does completing this task make me look smart?” and “Am I doing this task better than my friend?” (Wigfield & Cambria, 2010, p. 7). Both orientations are further classified based on their approach or avoidance nature (Scholer et al., 2019): mastery-approach (striving to learn), mastery-avoidance (striving to prevent loss of abilities and skills), performance-approach (striving for outperforming others), and performance-avoidance (striving for avoiding the display of incompetence).

Mastery-approach consistently predicts higher levels of effort and academic achievement, while the links between performance-approach and effort or academic achievement tend to be unstable and weak, especially among primary and secondary schoolers (Cellar et al., 2011; Gonida et al., 2009; Katz-Vago & Benita, 2024; Wigfield et al., 2015). Additionally, mastery-approach and effort are likely bidirectional or reciprocal (Katz-Vago & Benita, 2024; King et al., 2024). Individuals with a mastery-approach are more inclined to invest greater effort, driven by a desire to improve and learn. They believe that mastery and success result from dedication and hard work, viewing challenging tasks and difficult circumstances as opportunities to improve their understanding and build competence (Gonida et al., 2009; Guo et al., 2023). As they put forth more effort into their tasks, they are more likely to experience learning and progress. This sense of progress and mastery strengthens their belief in the value of mastery-approach, reinforcing the idea that their effort leads to improvement, progress, and competence, which in turn motivates them to continue tackling upcoming tasks with the same mindset (Guo et al., 2023; Phan, 2012).

Within the score of the present inquiry, we centered on expectancy and mastery-approach for the following considerations. Expectancy and mastery-approach are more closely linked to conceptual learning and academic achievement than value and other achievement goals, such as performance-approach (Barron & Hulleman, 2015; Trautwein et al., 2012). Our focus on expectancy is further substantiated by empirical evidence that highlights a “strong and almost complete overlap of constructs like self-concept of ability and self-efficacy” (Eccles & Wigfield, 2020, p. 4). In addition, contemporary

research on homework has been primarily stemmed from the instrumental and pragmatic necessities of understanding its function in enhancing student achievement (Fan et al., 2017; Xu & Corno, 2022).

While expectancy and mastery-approach are drawn from two distinct theoretical frameworks, they both fall within the realm of social-cognitive approaches to motivation, addressing the two interrelated questions, “Can I do it?” and “Do I want to do it?” (Bandhu et al., 2024, p. 12). It is not surprising that expectancy and mastery-approach are conceptually linked, as both emphasize personal growth and long-term goal pursuit – continuous self-improvement for mastery-approach and continuous perceived capability for expectancy. They also share a focus on developing competence and attaining task mastery through effort and learning (Eccles & Wigfield, 2020; Urdan & Kaplan, 2020). Additionally, both frameworks share “a common focus on action” rather than “the consequences of the action” (Urhahne & Wijnia, 2023, p. 22). Furthermore, examining the likely longitudinal linkages between expectancy, mastery-approach, effort, and achievement has significant theoretical and practical implications. This is especially in light of Eccles and Wigfield’s (2020) recent observation that “as long as we force all children to master the same set of intellectual skills, it is probably most adaptive for children and their teachers to believe quite strongly that people can improve their ability levels with well-targeted effort and good instruction” (p. 7).

Individuals with high expectancy for success are more prone to endorse a mastery-approach, as they feel confident in their capability to learn and improve, believe their efforts would lead to meaningful learning and mastery (instead of focusing solely on surpassing peers), and become more persistent and resilient when encountered with challenges (Diseth, 2011; Howard et al., 2021). Likewise, individuals with higher mastery-approach are more inclined to have higher expectancy, as continuous progress, enhanced competence, and the effort and resilience cultivated through mastery-approach can boost their expectations for future success (Abaszadeh et al., 2024; Kaplan & Flum, 2010). Moreover, the intrinsic motivation nurtured by mastery-approach can enhance expectancy, as learners gain confidence over time (Howard et al., 2021; King et al., 2024).

Additionally, expectancy and mastery-approach may interact or combine in a multiplicative fashion to shape achievement-related outcomes (Pekrun et al., 2006; Tamura et al., 2022). This implies that for a specific task, high expectancy may not compensate for low mastery approach, and high mastery-approach may not make up for low expectancy. In both scenarios, individuals are less inclined to succeed in the task. Conversely, when individuals have high levels of both expectancy and mastery-approach (i.e., aligning individuals’ expectancy with mastery-approach), achievement-related outcomes are likely to be maximized or especially high.

The reviewed literature collectively implies conceptual linkages among expectancy, mastery-approach, effort, and achievement. However, there appears to be a lack of research examining the longitudinal associations among these constructs. Our current study seeks to fill this gap in the literature.

1.2 Studies pertaining to homework

While no prior homework studies investigated the longitudinal associations among expectancy, mastery-approach, effort, and achievement, a number of studies pointed to likely linkages between these variables (Dettmers et al., 2010; Trautwein et al., 2006; Xu & Corno, 2022; Yang & Xu, 2018). In a study involving 3,483 ninth and tenth-grade students in German, Dettmers et al. (2010) reported there were positive zero-order associations between homework expectancy, value, and effort. Additionally, they found that homework expectancy (but not value) and effort were positively correlated with mathematics achievement, after accounting for other predictors (e.g., gender, SES, and prior achievement). Similarly, involving 1,799 tenth and eleventh-grade students in China, Xu (2017) reported a positive zero-order correlation between homework expectancy and value. In addition, Xu found that homework expectancy (but not value) was positively correlated with mathematics achievement.

Drawing from a sample of 1,439 Chinese students in seventh and ninth grade, Sun et al. (2019) found positive zero-correlations between mastery-approach, performance-approach, and homework effort. Moreover, they found that while mathematics achievement correlated positively with mastery-approach and performance-approach, the correlation with mastery-approach was stronger (medium effect size) compared to that with performance-approach (small effect size).

Xu and Corno (2022), involving 3,018 Chinese students in Grade 8, found positive bivariate correlations between homework expectancy, value, mastery-approach, performance-approach, and effort. Furthermore, they found that homework expectancy (but not value), mastery-approach (but not performance-approach), and effort correlated positively with mathematics achievement, after accounting for other predictors (e.g., parent education and prior knowledge). Based on these findings, Xu and Corno (2022) further suggest that “factors derived from both expectancy-value and goal-orientation theories of motivation have more explanatory power when taken together than when either is studied alone” (p. 553).

To summarize, this emerging line of homework studies indicates probable connections among expectancy, mastery-approach, effort, and achievement. In addition, it further supports the importance of focusing on these constructs within the homework context, as current research on expectancy-value and achievement goal theories (Barron & Hulleman, 2015; Eccles & Wigfield, 2020; Trautwein et al., 2012) indicates that homework expectancy and mastery-approach showed a stronger connection to academic achievement than value and performance-approach (Dettmers et al., 2010; Sun et al., 2019; Xu & Corno, 2022).

What remains unclear, however, is the intricate interplay of expectancy, mastery-approach, effort, and achievement over time. Furthermore, to our knowledge, no prior research has explored expectancy $\times$ mastery-approach interactions, particularly in the context of homework.

1.3 The present study

Expectancy-value and achievement goal theories, along with the related homework studies discussed above, suggest that expectancy, mastery-approach, effort, and achievement might shape each other over time (Abaszadeh et al., 2024; Cellar et al., 2011; Eccles & Wigfield, 2020; Katz-Vago & Benita, 2024; King et al., 2024; Tamura et al., 2022; Urdan & Kaplan, 2020; Xu & Corno, 2022). Hence, applying cross-lagged models is critical to study the longitudinal relationships among these constructs, since this approach incorporates autoregression to control for previous levels, thereby minimizing the risk of effect misestimation. Additionally, it permits empirical examination of influence directions over time, thereby strengthening causal inferences (Newsom, 2015; Selig & Little, 2012).

The present investigation intends to apply cross-lagged panel modeling to explore the longitudinal links among expectancy, mastery-approach, effort, and achievement involving middle schoolers over two measurement points in an academic year (Fig. 1). Grounded in expectancy-value theory (Eccles & Wigfield, 2020, 2023) and supporting research (Barron & Hulleman, 2015; Trautwein et al., 2012; Urhahne & Wijnia, 2023), we hypothesize reciprocal relationships between expectancy and effort, and between expectancy and achievement (Fig. 1). In addition, guided by achievement goal theory (Cellar et al., 2011; Urdan & Kaplan, 2020) and associated studies (Guo et al., 2023; Katz-Vago & Benita, 2024; King et al., 2024), we hypothesize reciprocal relationships between mastery-approach and effort, as well as between mastery-approach and achievement.

Furthermore, informed by expectancy-value and achievement goal theories, along with extant literature pertaining to these theories (Abaszadeh et al., 2024; Diseth, 2011; Howard et al., 2021; Kaplan & Flum, 2010; King et al., 2024), we hypothesize that expectancy and mastery-approach would be reciprocally related, as would effort and achievement. Finally, as expectancy and mastery-approach may combine or interact multiplicatively to shape academic outcomes (Pekrun et al., 2006; Tamura et al., 2022), we expect that the interaction between expectancy and mastery-approach would positively influence subsequent achievement.

2 Method

2.1 Participants

The research comprised 1,450 eighth-grade general education students (Average age = 14.1 ± 0.7 ; 48.6% female; 100% Han nationality) from forty-five classes in southeastern region of China, excluding students with special educational needs. In China's educational landscape, regular schools typically enroll students from a wide range of socioeconomic backgrounds, accommodating 85–90% of the student population with varying academic performance levels (Gao et al., 2011). In contrast, key schools, which have more resources and better-qualified teachers, are more selective, allocating 10–15% of placements to academically high-performance

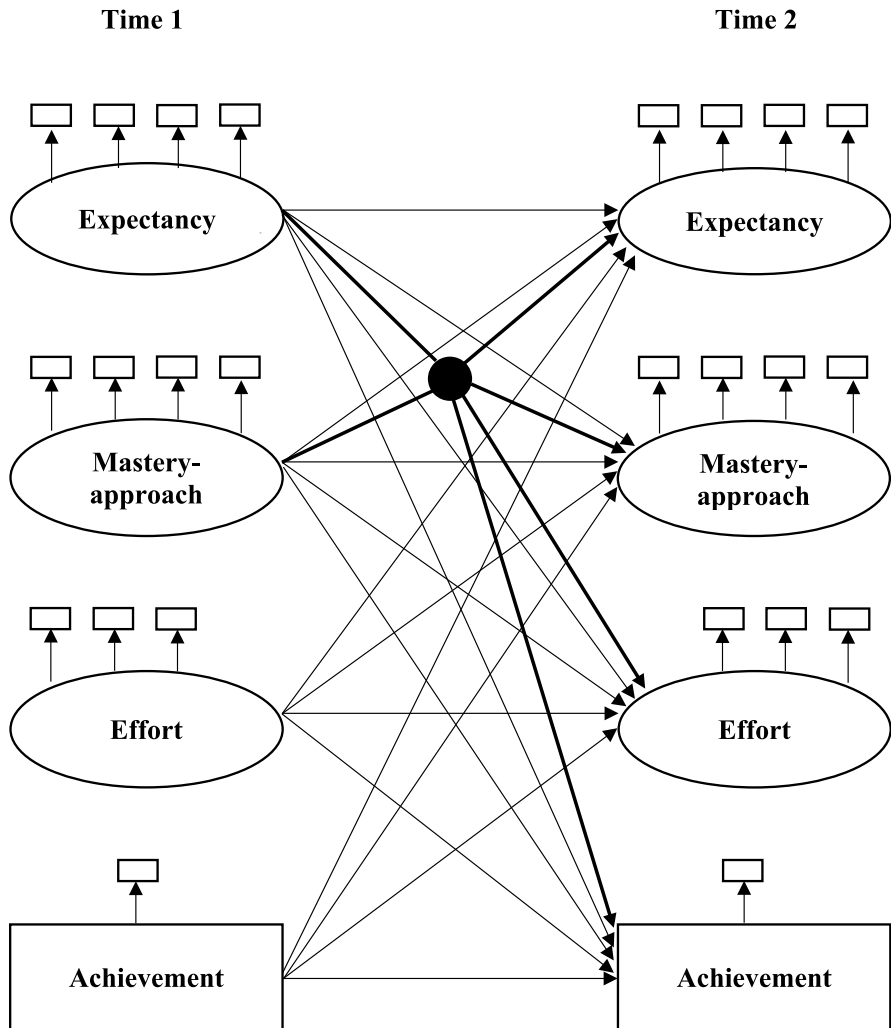


Fig. 1 Longitudinal relations among expectancy, mastery-approach, effort, achievement, and the interaction between expectancy and mastery-approach (depicted by the black circle and bold black paths)

students. To capture students from different socioeconomic backgrounds, the study drew participants from four regular public schools that permitted data collection.

For the participants' parents, the average years of maternal education were 9.9 ± 3.2 years, whereas paternal education averaged $10.4 \text{ years} \pm 3.1$. These figures are broadly consistent with the national working-age population average of 10.8 years in China (Huang & Lardy, 2021). Additionally, a positive correlation was found between the parents' education levels ($r = 0.70, p < 0.001$).

Guided by previous studies regarding suitable timeframes for investigating motivation, effort, and achievement (Marsh et al., 2016; Trautwein, 2007; Xu, 2018),

participants were assessed across two times: Time 1 (T1) in early November 2017 and Time 2 (T2) in the latter half of June 2018. At T1, 69.4% of students reported doing mathematics homework at least four days per week was 69.4% at Time 1, with a similar rate of 69.0% at Time 2. On average, participants spent 34.5 min per school day on mathematics homework at Time 1 and 35.3 min at Time 2. This practice aligned with findings from prior homework research in China (Xu & Corno, 2022).

2.2 Procedure

This study received ethical approval from a university ethics committee in China. All ethical guidelines were followed, including informed consent from parents. No compensation was provided for participation. Data collection took place during typical days, a research assistant asked teachers to leave their classroom to reduce any possible influence on student responses. An identifier was assigned to each participant to link mathematics achievement at two measurement points with corresponding measures (explained in the follow section). Following the completion of data collection, all identifiers were permanently removed.

2.3 Measures

2.3.1 Expectancy

At T1 and T2, the same 4-item scale measured homework expectancy (Xu, 2017; Xu & Corno, 2022). This scale tapped into participants' expectancy beliefs concerning mathematics homework (Table 1; e.g., the degree of confidence in the capabilities to accomplish mathematics assignments successfully; $\alpha/T1/T2 = 0.77/0.78$; $\omega/T1/T2 = 0.77/0.78$). Results from Confirmatory Factor Analysis (CFA) supported homework expectancy as a singular construct at T1 (CFI = 0.993; TLI = 0.980; SRMR = 0.016; RMSEA = 0.059) and T2 (CFI = 0.997; TLI = 0.991; SRMR = 0.012; RMSEA = 0.029). The reliability estimates aligned with prior research with Chinese students (Xu, 2023b; Xu & Corno, 2022; $0.79 \leq \alpha \leq 0.80$; $\omega = 0.80$).

2.3.2 Mastery-approach

At Time 1 and 2, the same 4-item scale evaluated mastery-approach (Sun et al., 2019; Xu & Corno, 2022). This scale tapped into the extent to which participants employed a mastery-approach to mathematics homework (e.g., to better understand homework tasks; $\alpha/T1/T2 = 0.84/0.85$; $\omega/T1/T2 = 0.84/0.85$). CFA results indicated that mastery-approach functioned as a unidimensional construct at T1 (CFI = 0.995; TLI = 0.984; SRMR = 0.014; RMSEA = 0.046) and T2 (CFI = 0.993; TLI = 0.978; SRMR = 0.013; RMSEA = 0.044). The reliability coefficients aligned with previous research with Chinese students (Sun et al., 2019; Xu & Corno, 2022; $\alpha = 0.86$; $\omega = 0.86$).

Table 1 Expectancy, mastery-approach, and effort items

Scales	Items	Kurtosis		Skewness		M (SD)		α/ω	
		T1	T2	T1	T2	T1	T2	T1	T2
Expectancy ^{a,c}	"If I don't understand something in math, I often think I'll never understand it."	0.370	-0.098	-0.919	-0.652	3.07 (0.66)	2.93 (0.68)	0.77/0.77	0.78/0.78
	"If I don't understand something in math, I'm at a complete loss and don't know how to catch up."	-0.635	-0.613	-0.343	-0.315				
	"Whether or not I do my math homework, I don't understand a thing in the lesson anyway."	0.408	-0.202	-1.074	-0.758				
	"I sometimes really dread math homework."	-0.860	-0.798	-0.252	-0.110				
Mastery-approach ^b	"I want to learn as much as possible in math homework."	-0.872	-0.990	-0.665	-0.458	4.94 (1.67)	4.71 (1.64)	0.84/0.84	0.85/0.85
	"I prefer math homework that really challenges me so I can learn new things."	-1.062	-1.003	-0.448	-0.299				
	"The most important thing for me is trying to understand math homework as thoroughly as possible."	-0.757	-0.902	-0.699	-0.466				
	"Understand math is important to me."	-0.437	-0.647	-0.847	-0.662				
Effort ^a	"I have recently been doing my math homework to the best of my ability."	0.843	0.753	-0.804	-0.761	3.19 (0.59)	3.09 (0.65)	0.75/0.75	0.84/0.84
	"I do my best on my math homework."	1.015	0.594	-0.847	-0.739				
	"I always try to finish my math homework."	0.619	0.567	-0.713	-0.685				

^a Rating: 1 (strongly disagree), 2 (disagree), 3 (agree), and 4 (strongly agree)^b Rating: 1 (not at all true of me) to 7 (very true of me)^c Items were reverse code

2.3.3 Effort

At Time 1 and 2, the same 3-item scale assessed homework effort, informed by prior work on homework effort (Trautwein et al., 2006; Xu & Corno, 2022). This scale evaluated how seriously participants worked on mathematics assignments ($\alpha/T1/T2 = 0.75/0.84$; $\omega/T1/T2 = 0.75/0.84$). As this scale contained only three items, a CFA would yield a just-identified model, rendering fit indices uninterpretable. Thus, we performed an exploratory factor analysis (EFA) to assess the underlying factor structure. EFA results supported the scale's unidimensionality, revealing (a) a one-factor solution that explained 66.4% of the variance at T1 and 75.4% at T2, and (b) strong factor loadings for all three items (ranging from 0.779 to 0.847 at T1 and 0.838 to 0.895 at T2). These reliability estimates aligned with prior studies with Chinese students (Xu, 2023a; Xu & Corno, 2022; $0.81 \leq \alpha \leq 0.82$; $0.82 \leq \omega \leq 0.83$).

2.3.4 Achievement

Mathematics achievement was assessed at T1 and T2, based on an exam aligned with national curriculum standards. The exam aimed to gauge students' grasp of mathematical understanding (e.g., triangle, fraction, and parallelogram; Ni et al., 2011). To enable linking test scores at T1 and T2, anchor items were included in the exam, which took 120 min to complete at each time point. The exam yielded an overall score, with coefficient alpha reliability values at 0.87 for T1 and 0.88 for T2.

2.4 Data analysis

We employed Mplus 8.11 to conduct data analyses. Expectancy, mastery-approach, and time management functioned as latent factors, with identical items used across two measurement points. The analysis incorporated cross-lagged and autoregressive pathways, correlations among expectancy, mastery-approach, effort, and achievement at both measurement points, as well as correlated uniqueness between identical indicators at both points. To appropriately handle the hierarchical structure (i.e., students within classes), Mplus's CLUSTER feature was used to adjust the standard errors.

At T1, 6.4% of the values were missing among 12 indicators (4 for expectancy, 4 for mastery-approach, 3 for effort, and 1 for achievement), whereas at T2, this percentage was 13.3%. Little's (1988) MCAR test evaluated whether missingness could be considered completely random. Findings supported for MCAR at T1 [$\chi^2(81) = 91.984$, $p = 0.190$], yet not at T2 [$\chi^2(145) = 204.794$, $p < 0.001$]. Consequently, we conducted our analyses applying MLR estimation with FIML, as FIML is recognized for producing unbiased estimates in the presence missing data (Marsh et al., 2016).

2.4.1 Invariance across two measurement points

In longitudinal studies, metric and scalar measurement invariance are crucial for ensuring the validity and comparability of measurements over time (Niepel et al., 2014; Van de Schoot et al., 2012). Metric invariance ensures that factor loadings

remain stable (i.e., consistent construct measurement), whereas scalar invariance ensures that observed score differences represent actual changes instead of measurement artifacts. Consequently, we assessed whether homework expectancy, mastery-approach, and effort captured the same constructs across Time 1 and 2. Mathematics achievement was not subjected to measurement invariance testing, since it was assessed using a single total score.

2.4.2 Gender invariance

Applying typically established practices for testing multiple group invariance (Hong et al., 2003; Yang & Xu, 2018), we examined a sequence of nested models over gender: configural, metric, correlated uniqueness, and scalar models. To test the invariance of path coefficients, we further constrained 16 pathways (4 horizontal and 12 cross) to be invariant across gender (Fig. 1).

2.4.3 Latent interaction

To examine the latent interaction between expectancy and mastery-approach, we used the latent moderated structural equations (LMS; Maslowsky et al., 2015). Whereas incorporating latent interaction does not impact model fit, the LMS cannot generate fitness indicators. Thus, we first assessed the model fit before introducing the expectancy $\times$ mastery-approach interaction (depicted by black circles and thick paths in Fig. 1).

2.4.4 Goodness of fit

Fit was assessed using multiple indices (Hu & Bentler, 1999): CFI ≥ 0.95 , TLI ≥ 0.95 , SRMR ≤ 0.08 , and RMSEA ≤ 0.06 . Additionally, with respect to multigroup invariance, we utilized the widely endorsed recommendation (Chen, 2007), indicating a preference for a more parsimonious model when $\Delta\text{CFI} \leq 0.010$ or $\Delta\text{RMESA} \leq 0.015$.

3 Results

Results are presented in two sections. Section 1 covers preliminary analysis, whereas Sect. 2 investigates longitudinal relationships among expectancy, mastery-approach, effort, and achievement from T1 and T2.

3.1 Preliminary analysis

Table 1 illustrates that the kurtosis and skewness of all observed variables fell within acceptable ranges (kurtosis: -1.062 to 1.015 ; skewness: -1.074 to -0.110). Additionally, Table 1 reports descriptive statistics and reliability estimates of expectancy, mastery-approach, and effort at T1 and T2.

For measurement invariance of expectancy, mastery-approach, and effort from T1 to T2, the configural invariance models yielded an excellent fit ($CFI \geq 0.990$ and $RMSEA \leq 0.030$). The transition to the metric invariance models resulted in very small changes in the fit indices ($\Delta CFI \leq 0.004$ and $\Delta RMSEA \leq 0.004$). The change from metric invariance to scalar invariance models was negligible ($\Delta CFI \leq 0.005$ and $\Delta RMSEA \leq 0.003$). These findings offered strong evidence in favor of scalar invariance for expectancy, mastery-approach, and effort, suggesting that these constructs were measured the same way across these two measurement points.

We examined two models as recommended by Maslowsky et al. (2015) – Model 0 and Model 1, which introduced latent interaction between expectancy and mastery-approach (Fig. 1). The data showed an excellent fit to Model 0 ($MLR\chi^2 = 413.433$; $df = 215$; $CFI = 0.981$; $TLI = 0.975$; $SRMR = 0.029$; $RMSEA = 0.025$; 90% CI = [0.022, 0.039]). This model accounted for 36.3% variance in T2 expectancy, 40.4% variance in T2 mastery-approach, 37.1% variance in T2 effort, and 73.3% variance in T2 achievement.

The fit of Model 0 versus Model 1 was compared using a log-likelihood ratio based on the log-likelihood values of Model 0 (−49,909.108) and Model 1 (−49,900.897). The log-likelihood difference was computed as $D = -2 \times [(-49,909.108) - (-49,900.897)] = 16.422$. The models had 109 (Model 0) and 113 (Model 1) free parameters, the significant log-likelihood ratio test ($p < 0.01$) indicated improved fit for Model 1 over Model 0. Overall, Model 1 accounted for 0.9% more variance in T2 achievement.

We then examined invariance across gender, including configural, metric, correlated uniqueness, scalar, and path coefficient invariance. The fit of path coefficient model was excellent ($CFI = 0.980$; $RMSEA = 0.024$), showing little differences from configural model ($\Delta CFI = 0.002$; $\Delta RMSEA = 0.001$).

Standardized factor loadings across two measurement points spanned from 0.560 to 0.807 for expectancy, 0.715 to 0.807 for mastery-approach, and 0.632 to 0.861 for effort. Within time points, the factor correlations were substantial between expectancy and mastery-approach (0.48 to 0.52), between mastery-approach and effort (0.50 to 0.51), and between expectancy and effort (0.38 to 0.43). Similarly, the correlations of achievement with expectancy (0.39 to 0.48), mastery-approach (0.41 to 0.51), and effort (0.35 to 0.40) were also considerable.

3.2 Longitudinal associations among expectancy, mastery-approach, effort, and achievement

3.2.1 Expectancy and achievement

Whereas expectancy was associated with achievement within each measurement point, we did not observe longitudinal relation between expectancy and achievement (Table 2 and Fig. 2). Expectedly, prior expectancy exerted the largest influence in shaping subsequent expectancy ($\beta = 0.460$, $p < 0.001$). Yet, higher T1 expectancy did not lead to higher T2 achievement ($\beta = 0.039$, $p = 0.126$), and higher T1 achievement did not result in higher T2 expectancy ($\beta = 0.076$, $p = 0.142$).

3.2.2 Mastery-approach and achievement

We observed no reciprocal relationship between mastery-approach and achievement. Initial mastery-approach exerted the greatest influence on subsequent mastery-approach ($\beta = 0.489$, $p < 0.001$) and showed a positive correlation with later achievement ($\beta = 0.118$, $p < 0.001$). Yet, initial achievement was not associated with later mastery-approach ($\beta = 0.072$, $p = 0.127$).

3.2.3 Effort and achievement

There was no longitudinal relation between effort and achievement. Whereas prior effort most strongly influenced subsequent effort ($\beta = 0.477$, $p < 0.001$), greater T1

Table 2 Standardized path coefficients among expectancy (EX), mastery-approach (MA), effort (EF), achievement (ACH), and EX $\times$ MA interaction

Dependent variable/ independent variable	Without EX $\times$ MA interaction		With EX $\times$ MA interaction	
	Path coefficient	Standard error	Path coefficient	Standard error
T2EX				
T1EX	0.459***	0.065	0.460***	0.062
T1MA	0.104*	0.043	0.104*	0.046
T1EX $\times$ T1MA			-0.005	0.043
T1EF	0.069	0.047	0.069	0.047
T1 ACH	0.078	0.053	0.076	0.052
T2MA				
T1EX	0.042	0.039	0.040	0.040
T1MA	0.493***	0.069	0.489***	0.067
T1EX $\times$ T1MA			-0.022	0.031
T1EF	0.131**	0.042	0.132**	0.042
T1 ACH	0.072	0.049	0.072	0.047
T2EF				
T1EX	0.051	0.050	0.052	0.050
T1MA	0.131**	0.046	0.129**	0.046
T1EX $\times$ T1MA			-0.013	0.037
T1EF	0.476***	0.072	0.477***	0.072
T1 ACH	0.055	0.043	0.054	0.042
T2 ACH				
T1EX	0.026	0.026	0.039	0.025
T1MA	0.100***	0.027	0.118***	0.028
T1EX $\times$ T1MA			0.063***	0.018
T1EF	0.026	0.023	0.021	0.023
T1 ACH	0.775***	0.029	0.762***	0.030

* $p < .05$. ** $p < .01$. *** $p < .001$

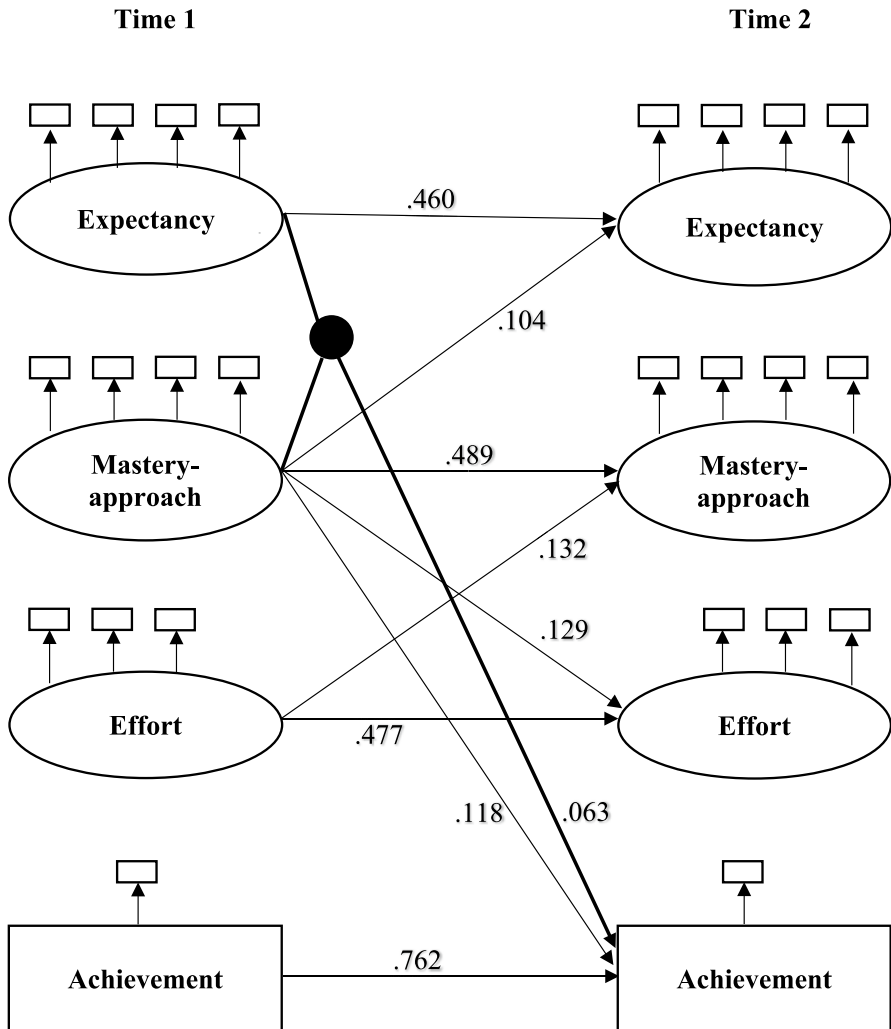


Fig. 2 Statistically significant pathways from T1 to T2 are displayed

effort did not lead to greater T2 achievement ($\beta = 0.021$, $p = 0.344$), nor did higher T1 achievement lead to higher T2 effort ($\beta = 0.054$, $p = 0.195$).

3.2.4 Expectancy and mastery-approach

We observed no reciprocal relationship between expectancy and mastery-approach. Whereas higher initial mastery-approach led to higher later expectancy ($\beta = 0.104$, $p = 0.023$), prior expectancy was unrelated to later mastery-approach ($\beta = 0.040$, $p = 0.316$).

3.2.5 Expectancy and effort

There was no longitudinal relationship between expectancy and effort. Prior expectancy was not related to subsequent effort ($\beta = 0.052$, $p = 0.304$), nor was prior effort related to subsequent expectancy ($\beta = 0.069$, $p = 0.140$).

3.2.6 Mastery-approach and effort

We observed reciprocal effects between mastery-approach and effort. Greater initial mastery-approach led to greater later effort ($\beta = 0.129$, $p = 0.005$), and greater initial effort led to greater later mastery-approach ($\beta = 0.132$, $p = 0.001$).

3.2.7 Expectancy $\times$ mastery-approach interaction

As displayed in Table 2, the expectancy $\times$ mastery-approach interaction had no significant effect on later expectancy ($\beta = -0.005$, $p = 0.910$), mastery-approach ($\beta = -0.022$, $p = 0.492$), or effort ($\beta = -0.013$, $p = 0.731$). Yet, the expectancy $\times$ mastery-approach interaction positively affected later achievement ($\beta = 0.063$, $p < 0.001$). Figure 3 illustrates the interaction effect on later achievement – T1 expectancy positively influenced T2 achievement when T1 mastery-approach was high, in addition to the positive influence of T1 mastery-approach on T2 achievement.

4 Discussion

Drawing on expectancy-value and achievement goal theories, this study examined the longitudinal relations among expectancy, mastery-approach, homework effort, and achievement. We investigated several hypotheses, some building on prior research, whereas others providing fresh and intriguing insights.

4.1 Reciprocal influences of mastery-approach and effort

We observed positive reciprocal influences of mastery-approach and homework effort (Fig. 3): higher initial mastery-approach led to higher later homework effort, and higher initial homework effort resulted in higher later mastery-approach. Whereas these results are consistent with achievement goal theory (Cellar et al., 2011; Urdan & Kaplan, 2020) and extant literature (Gonida et al., 2009; Guo et al., 2023; Katz-Vago & Benita, 2024; King et al., 2024), the current investigation extends existing literature by applying cross-lagged models to examine the longitudinal associations between mastery-approach and effort.

Additionally, this investigation advanced our understanding by revealing that the path from prior effort to subsequent mastery-approach ($\beta = 0.132$) was almost identical to the path from prior mastery-approach to subsequent effort ($\beta = 0.129$), with both classified as large effect sizes per the newly proposed guidelines cross-lagged

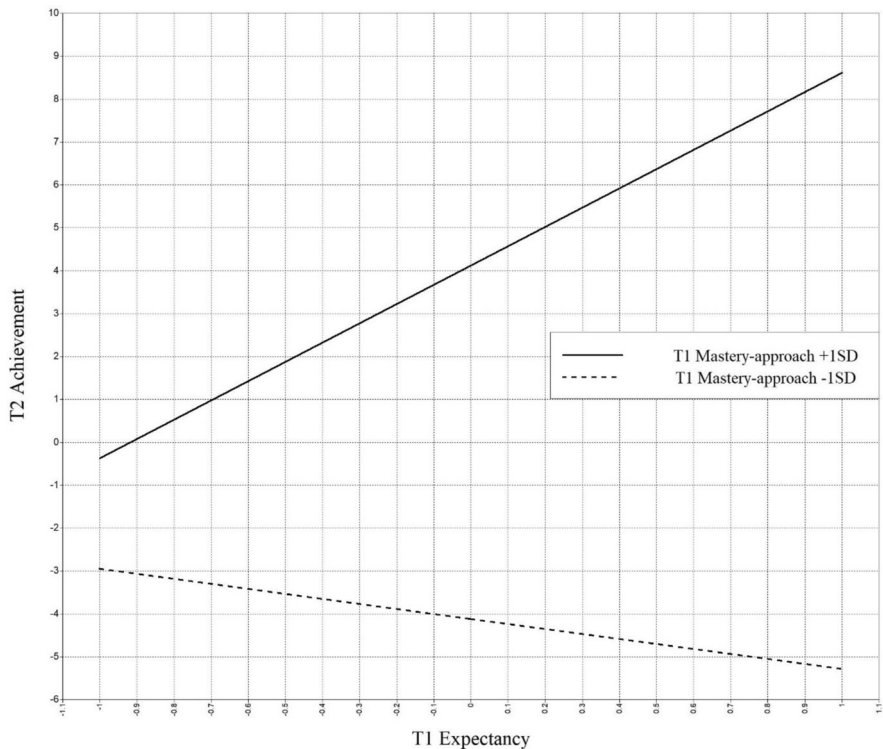


Fig. 3 Interaction between T1 Expectancy and T1 Mastery-approach predicts T2 Achievement – the influence of T1 Expectancy on T2 Achievement at 1SD below and above the mean of T1 Mastery-approach

effect size guidelines (0.12 for large; Orth et al., 2024). These findings indicated that, as expected, prior mastery-approach (or effort) played an influential role in subsequent mastery-approach (or effort). They further indicated that the influence of prior mastery-approach in enhancing subsequent homework effort was almost exactly matched by the influence of prior homework effort in boosting subsequent mastery-approach. In other words, homework effort could shape mastery-approach as much as mastery-approach could shape homework effort, highlighting the balanced interdependence and equal potency of homework effort and mastery-approach in the homework process.

4.2 The role of mastery-approach on expectancy

An asymmetrical influence was observed between mastery-approach and expectancy (Fig. 3): greater initial mastery-approach led to greater later expectancy, yet initial expectancy did not influence later mastery-approach. One plausible explanation is that expectancy of success is a proximal explanatory factor shaped by individuals' goals such as mastery-approach (Eccles & Wigfield, 2020; Urhahne & Wijnia,

2023). Prior mastery-approach may lead students to approach homework with an intrinsically motivated mindset focused on understanding and mastery, which may result in higher subsequent expectancy as students gain more confidence in their ability to succeed in subsequent homework tasks. On the other hand, mastery-approach is more intrinsically driven by an inherent desire to learn and develop competence, instead of being contingent on more external factors such as expectancies of success. Hence, initial expectancies may not directly affect intrinsic motivational tendencies such as mastery-approach, and the influence of initial expectancy on later mastery-approach may be less apparent.

Another explanation is that the influence of initial expectancy on later mastery-approach may be moderated by cultural differences. China, as with many East Asian nations affected by Confucian cultural legacy, strongly emphasizes mastery-approach (i.e., learning, understanding, and self-improvement) and intrinsic motivation specifically in subjects such as mathematics (e.g., the importance of mathematical proficiency; Chen & Wong, 2015) and in learning contexts like homework (Chen & Stevenson, 1989; Xu et al., 2016). The cultural emphasis on mastery-approach often motivates Chinese students more than expectations of success.

Moreover, in Chinese culture, effort and resilience (instead of innate ability) are viewed as the key to educational success (Cui et al., 2023; Li, 2001; Xu, 2018). Chinese students are consistently encouraged and internalized to persist through challenges and obstacles in their academic pursuits. Thus, their mastery-approach is likely coupled with a strong focus on effort, potentially diminishing the effect of initial expectancy on later mastery-approach. This explanation is substantiated by our findings, which showed reciprocal relationships between effort and mastery-approach, whereas there was no longitudinal connection between effort and expectancy.

Another related explanation is that Chinese culture is characterized by “long-term orientation” (Hofstede, 1980; Wang, 2018). Chinese students are likely to adopt a more long-term perspective on their academic pursuits and goals, viewing education as a continuous process of learning and sustained effort. They perceive setbacks as opportunities for growth or self-improvement rather than reflections of their ability or potential success. This long-term perspective may further mitigate the influence of prior expectancy on later mastery-approach.

4.3 The role of mastery-approach on achievement

We found an asymmetrical influence between mastery-approach and achievement (Fig. 3): greater T1 mastery-approach led to greater T2 achievement, but T1 achievement did not influence T2 mastery-approach. Our result relating to the positive influence of T1 mastery-approach on T2 achievement aligns with the literature on achievement goal (Katz-Vago & Benita, 2024; Senko et al., 2011) as well as related homework studies regarding the influential role of mastery-approach on student achievement (Sun et al., 2019; Xu & Corno, 2022).

Mastery-approach is driven by a focus on learning and self-improvement. This internal drive prompts students to approach homework assignments with

greater persistence and determination. Rather than perceiving challenges as indicators of their capabilities, they view obstacles and setbacks as opportunities for growth and development. Such a mindset promotes learning outcomes such as higher subsequent achievement. In other words, mastery-approach may be shaped more by students' internal drive for learning and self-improvement than by their abilities or prior academic achievement. As such, it seems logical that higher prior mastery-approach results in higher later achievement, not vice versa.

Another related explanation is that students with mastery-approach tend to have a growth mindset, believing that abilities can be developed through effort and learning (Guo et al., 2023; Payne et al., 2007). This explanation is substantiated by results from our current study that master-approach and effort were reciprocally related. This mindset may buffer against the potential influence of lower achievement, leading students to sustain the focus on mastery regardless of their performance. As a result, the influence of initial achievement on later mastery-approach is likely to be less pronounced.

4.4 Expectancy $\times$ mastery-approach interaction

The present study revealed that the expectancy $\times$ mastery-approach interaction positively influenced T2 achievement, indicating that prior expectancy exerted a positive influence on T2 achievement when T1 mastery-approach was high (the multiplicative effect). While our result aligns with the propositions in existing literature (Pekrun et al., 2006; Tamura et al., 2022), the current study makes a significant advancement, by extending previous research on expectancy and mastery-approach through the use of cross-lagged panel models in the context of homework.

Our finding indicates that the interaction between expectancy and mastery-approach generate a powerful motivational force. This finding is particularly noteworthy within the context of our study. As discussed earlier, prior expectancy did not influence subsequent achievement in the same way mastery-approach did, as Chinese students are more inclined to be motivated by the cultural emphasis on mastery-approach rather than their expectancies of success. Yet, when students have higher expectancy combined with a strong desire to master the content (i.e., both confident and strategically focused on learning), they are more inclined to achieve higher academic outcomes.

This suggests that mastery-approach can magnify the influence of higher expectancy by offering a clear and more motivating pathway for students to channel their confidence and propel them to exert greater effort aimed at learning, understanding, growth, and self-improvement, leading to higher levels of achievement over time. In addition, students with higher mastery-approach and expectancy are better positioned to persist through challenges and setbacks, by viewing difficulties and obstacles as opportunities to learn and improve, rather than as threats to their ability or self-worth.

4.5 Strengths, limitations, and implications for further research

This study represents a significant advancement over previous research by integrating expectancy-value and achievement goal theories and examining the longitudinal interplay among expectancy, mastery-approach, effort, and achievement. Notably, as no prior studies, to our knowledge, have investigated the dynamic relationships among these constructs over time. Our investigation expands on prior research by revealing (a) the reciprocal influence between mastery-approach and effort, (b) the asymmetrical effects of mastery-approach on expectancy and achievement, and (c) the latent interaction between expectancy and mastery-approach on subsequent achievement. In addition, the invariance of path coefficients across gender found in our study implies that these results are applicable to both girls and boys. Lastly, based on the recently introduced benchmarks for cross-lagged effect sizes (effect sizes of 0.12, 0.07, and 0.03 denoting large, medium, and small effects; Orth et al., 2024), the effect sizes observed in our study range from approaching medium (0.063 for T1 expectancy $\times$ T1 mastery-approach to T2 achievement) to large (0.132 for T1 effort to T2 mastery-approach).

Several limitations warrant acknowledgement. First, while our study included standardized mathematics achievement at two measurement points, we measured expectancy, mastery-approach, and effort relied on self-reports. As self-reports may result in inflationary bias because of shared method effects (e.g., inflated relationships among constructs; Marsh et al., 2016; Xu, 2025b), it would be important to incorporate multiple data sources in future research (e.g., teacher report and observation). Yet, as noted by other researchers (e.g., Marsh et al., 2016), it is important to recognize that given our emphasis on student motivation, self-report remains the most prevalent and suitable measure for assessing students' perceptions of confidence, goal orientation, and effort, despite the potential for alternative sources to offer different insights.

Furthermore, we targeted middle school mathematics homework in our study. Therefore, it would be beneficial to conduct future research involving elementary or high schoolers, given that (a) meta-analyses indicated that the association between homework and achievement was influenced by grade level (Fan et al., 2017), and (b) expectancy and mastery-approach tend to evolve over time (Eccles & Wigfield, 2023; Tuominen et al., 2020; Wigfield et al., 2015). Additionally, it would be important to pursue this line of research in other subjects, as previous studies often focus on mathematics (Dettmers et al., 2010; Xu, 2024c), and as mathematics often requires more effort from students compared with other subjects (Marsh et al., 2016; Silinskas & Kikas, 2019; Xu, 2022).

Finally, our investigation involved Chinese students. Hence, it is vital to carry on this line of research in different cultural settings, as (a) cultural differences may affect the longitudinal linkages among expectancy, mastery-approach, effort, and achievement (e.g., the emphasis on mastery-approach, effort, resilience; Chen & Wong, 2015; Wigfield et al., 2015; Xu, 2023b), and (b) as our investigation further suggests that some results (e.g., the asymmetrical relationship between expectancy and mastery-approach) might be affected by cultural differences.

4.6 Practical implications

As the current investigation represents the first attempt to use cross-lagged models to study the longitudinal linkages among expectancy, mastery-approach, effort, and achievement, more research is needed before making conclusive recommendations for practice.

Our results concerning reciprocal influence of mastery-approach and homework effort suggest the critical importance of promoting mastery-approach and homework effort simultaneously. Aligning with previous homework research on mastery-approach (Sun et al., 2019; Xu & Corno, 2022), it would be advantageous for teachers to emphasize mastery-approach toward homework tasks. Pivotal to this approach would be spotlighting homework's function in promoting learning and growth, while guiding students to recognize its importance as a crucial means to better understand academic material and close critical gaps in their academic experience. This might entail assigning homework tasks that are appropriately challenging and relevant to students' learning objectives, offering constructive and timely feedback that centers on improvement, progress, and growth instead of the final outcome, and creating an atmosphere where mistakes or errors are viewed as opportunities for learning.

This emphasis on mastery-approach is especially important, as our findings further indicated that prior higher mastery-approach resulted in higher later expectancy and achievement (not the other way around). Additionally, given our result that the latent expectancy $\times$ mastery-approach interaction positively influenced subsequent achievement, there is a need to promote homework expectancy combined with a strong desire to master the content. It would be beneficial for teachers, for example, to design homework assignments that are challenging yet attainable, to provide explicit guidance and ample opportunities for mastery experience and build their confidence, and to encourage students to revise assignments for deeper understanding. Furthermore, teachers may offer extra support to students with lower homework expectancy, while encouraging those with higher expectancy and mastery-approach to share their strategies and insights, thereby serving as role models for their peers.

Finally, given that homework effort can influence mastery-approach as much as mastery-approach can influence homework effort, it would be critically important for teachers to promote sustained effort in homework. This may entail cultivating an environment that recognizes and celebrates diligence and resilience, that encourages risk-taking and learning from mistakes, and that prompts students to reflect on how their effort attributes to their learning and growth.

Declarations

Conflict of interest The author has no relevant financial or non-financial interests to disclose.

Ethical approval The study (involving human participants; informed consent) was approved the Institutional Review Board, University of Macau (MYRG2017-00122-FED).

References

- Abaszadeh, H., Amani, M., & Pordanjani, T. R. (2024). The relationship between motivational-cognitive variables, academic self-efficacy of students mediated by parent's educational expectations, parent-child interaction, and teacher-student interaction. *Learning and Motivation*, 86, 101983. <https://doi.org/10.1016/j.lmot.2024.101983>
- Bandhu, D., Mohan, M. M., Nittala, N. A. P., Jadhav, P., Bhadauria, A., & Saxena, K. K. (2024). Theories of motivation: A comprehensive analysis of human behavior drivers. *Acta Psychologica*, 244, 104177. <https://doi.org/10.1016/j.actpsy.2024.104177>
- Barron, K. E., & Hulleman, C. S. (2015). The expectancy-value-cost model of motivation. In J. D. Wright (Ed.), *International encyclopedia of the social and behavioral sciences* (2nd ed., Vol. 8, pp. 503–509). Elsevier.
- Cellar, D. F., Stuhlmacher, A. F., Young, S. K., Fisher, D. M., Adair, C. K., Haynes, S., Twichell, E., Arnold, K. A., Royer, K., Denning, B. L., & Riestler, D. (2011). Trait goal orientation, self-regulation, and performance: A meta-analysis. *Journal of Business and Psychology*, 26(4), 467–483. <https://doi.org/10.1007/s10869-010-9201-6>
- Chen, C., & Stevenson, H. W. (1989). Homework: A cross-cultural examination. *Child Development*, 60, 551–561. <https://doi.org/10.2307/1130721>
- Chen, F. F. (2007). Sensitivity of goodness of fit indices to lack of measurement invariance. *Structural Equation Modeling*, 14, 464–504. <https://doi.org/10.1080/10705510701301834>
- Chen, W. W., & Wong, Y. L. (2015). Chinese mindset: Theories of intelligence, goal orientation and academic achievement in Hong Kong students. *Educational Psychology*, 35, 714–725. <https://doi.org/10.1080/01443410.2014.893559>
- Corno, L., & Xu, J. (2004). Doing homework as the job of childhood. *Theory into Practice*, 43(3), 227–233. https://doi.org/10.1207/s15430421tip4303_9
- Cui, T., Wang, C., & Xu, J. (2023). Validation of Academic Resilience Scales adapted in a collectivistic culture. *Frontiers in Psychology*, 14, 1114285. <https://doi.org/10.3389/fpsyg.2023.1114285>
- Dettmers, S., Trautwein, U., Lüdtke, O., Kunter, M., & Baumert, J. (2010). Homework works if homework quality is high: Using multilevel modeling to predict the development of achievement in mathematics. *Journal of Educational Psychology*, 102, 467–482. <https://doi.org/10.1037/a0018453>
- Diseth, Å. (2011). Self-efficacy, goal orientations and learning strategies as mediators between preceding and subsequent academic achievement. *Learning and Individual Differences*, 21(2), 191–195. <https://doi.org/10.1016/j.lindif.2011.01.003>
- Eccles, J. S., Adler, T. F., Futterman, R., Goff, S. B., Kaczala, C. M., Meece, J. L., & Midgley, C. (1983). Expectancies, values, and academic behaviors. In J. T. Spence (Ed.), *Achievement and achievement motivation* (pp. 75–146). W. H. Freeman.
- Eccles, J. S., & Wigfield, A. (2020). From expectancy-value theory to situated expectancy-value theory: A developmental, social cognitive, and sociocultural perspective on motivation. *Contemporary Educational Psychology*, 61, 101859. <https://doi.org/10.1016/j.cedpsych.2020.101859>
- Eccles, J. S., & Wigfield, A. (2023). Expectancy-value theory to situated expectancy-value theory: Reflections on the legacy of 40+ years of working together. *Motivation Science*, 9(1), 1–12. <https://doi.org/10.1037/mot0000275>
- Fan, H., Xu, J., Cai, Z., He, J., & Fan, X. (2017). Homework and students' achievement in math and science: A 30-year meta-analysis, 1986–2015. *Educational Research Review*, 20, 35–54. <https://doi.org/10.1016/j.edurev.2016.11.003>
- Fernandez-Alonso, R., Suarez-Alvarez, J., & Muniz, J. (2015). Adolescents' homework performance in mathematics and science: Personal factors and teaching practices. *Journal of Educational Psychology*, 107(4), 1075–1085. <https://doi.org/10.1037/edu0000032>
- Gao, X., Barkhuizen, G., & Chow, A. (2011). 'Nowadays, teachers are relatively obedient': Understanding primary school English teachers' conceptions of and drives for research in China. *Language Teaching Research*, 15, 61–81. <https://doi.org/10.1177/1362168810383344>
- Gonida, E. N., Voulala, K., & Kiosseoglou, G. (2009). Students' achievement goal orientations and their behavioural and emotional engagement: Co-examining the role of perceived school goal structures and parent goals during adolescence. *Learning and Individual Differences*, 19, 53–60. <https://doi.org/10.1016/j.lindif.2008.04.002>
- Guo, J., Hu, X., Elliot, A. J., Marsh, H. W., Murayama, K., Basarkod, G., Parker, P. D., & Dicke, T. (2023). Mastery-approach goals: A large-scale cross-cultural analysis of antecedents and

- consequences. *Journal of Personality and Social Psychology*, 125(2), 397–420. <https://doi.org/10.1037/pspp0000436>
- Hofstede, G. (1980). *Culture's consequences: International differences in work-related values*. Sage.
- Hong, S., Malik, M. L., & Lee, M.-K. (2003). Testing configural, metric, scalar, and latent mean invariance across genders in sociotropy and autonomy using a non-Western sample. *Educational and Psychological Measurement*, 63(4), 636–654. <https://doi.org/10.1177/0013164403251332>
- Howard, J. L., Bureau, J., Guay, F., Chong, J. X. Y., & Ryan, R. M. (2021). Student motivation and associated outcomes: A meta-analysis from self-determination theory. *Perspectives on Psychological Science*, 16(6), 1300–1323. <https://doi.org/10.1177/1745691620966789>
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6, 1–55. <https://doi.org/10.1080/10705519909540118>
- Huang, T., & Lardy, N. (2021). Opinion: How China can make its shrinking workforce more educated. Caixin Global. Available online: <https://www.caixinglobal.com/2021-03-24/opinion-how-china-can-make-its-shrinking-workforce-more-educated-101679922.html#:~:text=Aside%20from%20rolling%20out%20a,year%20of%20education%20per%20year> (Accessed on 12 December 2024).
- Kaplan, A., & Flum, H. (2010). Achievement goal orientations and identity formation styles. *Educational Research Review*, 5(1), 50–67. <https://doi.org/10.1016/j.edurev.2009.06.004>
- Katz-Vago, I., & Benita, M. (2024). Mastery-approach and performance-approach goals predict distinct outcomes during personal academic goal pursuit. *British Journal of Educational Psychology*, 94(2), 309–327. <https://doi.org/10.1111/bjep.12645>
- Kennedy, T. J., & Odell, M. R. L. (2023). STEM education as a meta-discipline. In B. Akpan, B. Cavas, & T. Kennedy (Eds.), *Contemporary issues in science and technology education* (pp. 37–51). Springer Nature.
- King, R. B., Wang, F., Leung, S. O., & Elliot, A. (2024). Socio-economic status, mastery-approach goals and learning-related outcomes: Mediation and moderation. *British Journal of Educational Psychology*, 94(2), 499–517. <https://doi.org/10.1111/bjep.12660>
- Li, J. (2001). Chinese conceptualization of learning. *Ethos*, 29, 111–137. <https://doi.org/10.1525/eth.2001.29.2.111>
- Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83, 1198–1202. <https://doi.org/10.1080/01621459.1988.10478722>
- Marsh, H. W., Pekrun, R., Lichtenfeld, S., Guo, J., Arens, A. K., & Murayama, K. (2016). Breaking the double-edged sword of effort/trying hard: Developmental equilibrium and longitudinal relations among effort, achievement, and academic self-concept. *Developmental Psychology*, 52, 1273–1290. <https://doi.org/10.1037/dev0000146>
- Maslowsky, J., Jager, J., & Hemken, D. (2015). Estimating and interpreting latent variable interactions: A tutorial for applying the latent moderated structural equations method. *International Journal of Behavioral Development*, 39, 87–96. <https://doi.org/10.1177/0165025414552301>
- Newsom, J. T. (2015). *Longitudinal structural equation modeling: A comprehensive introduction*. Routledge.
- Ni, Y., Li, Q., Li, X., & Zhang, Z. H. (2011). Influence of curriculum reform: An analysis of student mathematics achievement in Mainland China. *International Journal of Educational Research*, 50, 100–116. <https://doi.org/10.1016/j.ijer.2011.06.005>
- Niepel, C., Brunner, M., & Preckel, F. (2014). The longitudinal interplay of students' academic self-concepts and achievements within and across domains: Replicating and extending the reciprocal internal/external frame of reference model. *Journal of Educational Psychology*, 106, 1170–1191. <https://doi.org/10.1037/a0036307>
- Orth, U., Meier, L. L., Bühler, J. L., Dapp, L. C., Krauss, S., Messerli, D., & Robins, R. W. (2024). Effect size guidelines for cross-lagged effects. *Psychological Methods*, 29(2), 421–433. <https://doi.org/10.1037/met0000499>
- Payne, S. C., Youngcourt, S. S., & Beaubien, J. M. (2007). A meta-analytic examination of the goal orientation nomological net. *Journal of Applied Psychology*, 92(1), 128–150. <https://doi.org/10.1037/0021-9010.92.1.128>
- Pekrun, R., Elliot, A. J., & Maier, M. A. (2006). Achievement goals and discrete achievement emotions: A theoretical model and prospective test. *Journal of Educational Psychology*, 98(3), 583–597. <https://doi.org/10.1037/0022-0663.98.3.583>

- Phan, H. P. (2012). Prior academic achievement, effort, and achievement goal orientations: A longitudinal examination. *Journal of Educational and Developmental Psychology*, 2(2), 57–71. <https://doi.org/10.5539/jedp.v2n2p57>
- Scholer, A. A., Cornwell, J. F., & Higgins, E. T. (2019). Should we approach approach and avoid avoidance? An inquiry from different levels. *Psychological Inquiry*, 30(3), 111–124. <https://doi.org/10.1080/1047840X.2019.1643667>
- Selig, J. P., & Little, T. D. (2012). Autoregressive and cross-lagged panel analysis for longitudinal data. In B. Laursen, T. D. Little, & N. A. Card (Eds.), *Handbook of Developmental Research Methods* (pp. 265–278). Gilford Press.
- Senko, C., Hulleman, C. S., & Harackiewicz, J. M. (2011). Achievement goal theory at the crossroads: Old controversies, current challenges, and new directions. *Educational Psychologist*, 46, 26–47. <https://doi.org/10.1080/00461520.2011.538646>
- Silinskas, G., & Kikas, E. (2019). Parental involvement in math homework: Links to children's performance and motivation. *Scandinavian Journal of Educational Research*, 63(1), 17–37. <https://doi.org/10.1080/00313831.2017.1324901>
- Sun, M., Du, J., Xu, J., & Liu, F. (2019). Homework Goal Orientation Scale: Measurement invariance and latent mean differences across gender and grade level. *Psychology in the Schools*, 56(3), 465–477. <https://doi.org/10.1002/pits.22206>
- Tamura, A., Ishii, R., Yagi, A., Fukuzumi, N., Hatano, A., Sakaki, M., Tanaka, A., & Murayama, K. (2022). Exploring the within-person contemporaneous network of motivational engagement. *Learning and Instruction*, 81, 1–12. <https://doi.org/10.1016/j.learninstruc.2022.101649>
- Trautwein, U. (2007). The homework-achievement relation reconsidered: Differentiating homework time, homework frequency, and homework effort. *Learning and Instruction*, 17, 372–388. <https://doi.org/10.1016/j.learninstruc.2007.02.009>
- Trautwein, U., Ludtke, O., Schnyder, I., & Niggli, A. (2006). Predicting homework effort: Support for a domain-specific, multilevel homework model. *Journal of Educational Psychology*, 98, 438–456. <https://doi.org/10.1037/0022-0663.98.2.438>
- Trautwein, U., Marsh, H. W., Nagengast, B., Lüdtke, O., Nagy, G., & Jonkmann, K. (2012). Probing for the multiplicative term in modern expectancy-value theory: A latent interaction modeling study. *Journal of Educational Psychology*, 104, 763–777. <https://doi.org/10.1037/0022-0663.98.2.438>
- Tuominen, H., Niemivirta, M., Lonka, K., & Salmela-Aro, K. (2020). Motivation across a transition: Changes in achievement goal orientations and academic well-being from elementary to secondary school. *Learning and Individual Differences*, 79, 101854. <https://doi.org/10.1016/j.lindif.2020.101854>
- Urdan, T., & Kaplan, A. (2020). The origins, evolution, and future directions of achievement goal theory. *Contemporary Educational Psychology*, 61, 101862. <https://doi.org/10.1016/j.cedpsych.2020.101862>
- Urhahne, D., & Wijnia, L. (2023). Theories of motivation in education: An integrative framework. *Educational Psychology Review*, 35(2), 45. <https://doi.org/10.1007/s10648-023-09767-9>
- Van de Schoot, R., Lugtig, P., & Hox, J. (2012). A checklist for testing measurement invariance. *European Journal of Developmental Psychology*, 9, 486–492. <https://doi.org/10.1080/17405629.2012.686740>
- Wang, I. K. H. (2018). Long-term Chinese students' transitional experiences in UK higher education: A particular focus on their academic adjustment. *International Journal of Teaching and Learning in Higher Education*, 30(1), 12–25.
- Wang, X. L. (2013). Why students choose STEM majors: Motivation, high school learning, and postsecondary context of support. *American Educational Research Journal*, 50(5), 1081–1121. <https://doi.org/10.3102/0002831213488622>
- Wigfield, A., & Cambria, J. (2010). Students' achievement values, goal orientations, and interest: Definitions, development, and relations to achievement outcomes. *Developmental Review*, 30, 1–35. <https://doi.org/10.1016/j.dr.2009.12.001>
- Wigfield, A., Eccles, J. S., Fredricks, J. A., Simpkins, S., Roeser, R. W., & Schiefele, U. (2015). Development of achievement motivation and engagement. In M. E. Lamb & R. M. Lerner (Eds.), *Handbook of child psychology and developmental science, Socioemotional processes* (7th ed., pp. 657–700). Wiley.
- Xiang, P., Liu, J., Li, W., & Guan, J. (2022). Longitudinal stability and change in young students' mastery-approach goals for running. *Perceptual and Motor Skills*, 129(5), 1581–1598. <https://doi.org/10.1177/00315125221118469>

- Xu, J. (2017). Homework Expectancy Value Scale for high school students: Measurement invariance and latent mean differences across gender and grade level. *Learning and Individual Differences*, 60, 10–17. <https://doi.org/10.1016/j.lindif.2017.10.003>
- Xu, J. (2018). Reciprocal effects of homework self-concept, interest, effort, and math achievement. *Contemporary Educational Psychology*, 55, 42–52. <https://doi.org/10.1016/j.cedpsych.2018.09.002>
- Xu, J. (2022). Investigating factors that influence math homework purposes: A multilevel analysis. *Journal of Experimental Education*, 90(4), 862–883. <https://doi.org/10.1080/00220973.2020.1810604>
- Xu, J. (2023a). Student-perceived parental help with homework: Identifying student profiles and their relations with homework effort, procrastination, and achievement. *Learning and Individual Differences*, 104, 102299. <https://doi.org/10.1016/j.lindif.2023.102299>
- Xu, J. (2023). Taking a person-centered approach to student homework motivation: Combining achievement goal and expectancy-value theories. *Current Psychology*, 42, 29893–29904. <https://doi.org/10.1007/s12144-022-04044-4>
- Xu, J. (2024a). Homework time management: Do teacher and parent autonomy support matter? *Social Psychology of Education*, 27, 1947–1966. <https://doi.org/10.1007/s11218-024-09891-6>
- Xu, J. (2024b). Investigating factors influencing students' regulation of homework emotion: Integrating multiple theoretical perspectives. *Metacognition and Learning*, 19(3), 999–1024. <https://doi.org/10.1007/s11409-024-09395-w>
- Xu, J. (2024c). Student-perceived teacher and parent homework involvement: Exploring latent profiles and links to homework behavior and achievement. *Learning and Individual Differences*, 109, 102403. <https://doi.org/10.1016/j.lindif.2023.102403>
- Xu, J. (2025a). Longitudinal associations between autonomy support, help seeking, and achievement. *Learning and Individual Differences*, 117, 102583. <https://doi.org/10.1016/j.lindif.2024.102583>
- Xu, J. (2025b). Online homework time, time management, and procrastination: A person-centered approach. *Computers & Education*, 234, 105341. <https://doi.org/10.1016/j.compedu.2025.105341>
- Xu, J., & Corno, L. (2022). Extending a model of homework: A multilevel analysis with Chinese middle school students. *Metacognition and Learning*, 17(2), 531–563. <https://doi.org/10.1007/s11409-022-09296-w>
- Xu, J., Guo, S., Feng, Y., Ma, Y., Zhang, Y., Núñez, J. C., & Fan, H. (2024). Parental homework involvement and students' achievement: A three-level meta-analysis. *Psicothema*, 36(1), 1–14. <https://doi.org/10.7334/psicothema2023.92>
- Xu, J., Yuan, R., Xu, B., & Xu, M. (2016). Modeling students' interest in math homework. *Journal of Educational Research*, 109(2), 148–158. <https://doi.org/10.1080/00220671.2014.928252>
- Yang, F., & Xu, J. (2018). Homework Expectancy Value Scale: Measurement invariance and latent mean differences across gender. *Journal of Psychoeducational Assessment*, 36, 863–868. <https://doi.org/10.1177/0734282917714905>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.